



Sri Lanka Institute of Information Technology

Faculty of Computing

SE2030 - Software Engineering | Lab Sheet 06 - Software Testing

Test Case Document

Project: Web-Based Library Management System

Group ID: Group 02 | Year 2, Semester 1 (2026)

GitHub: <https://github.com/IT21060144/SE2030-Group2-LibrarySystem>

Group Members

Student ID	Name	Assigned Function
IT21047756	Weerasekara W.B.M.K.D	Return Management
IT21054372	Samarasinghe P.D.P	Book Management
IT21060144	Rathnayake R.M.I	Reservation Management
IT21077524	Pathirana A.P.C.E	User Management
IT21101274	Dolawatta T.Y.R	Fine Management
IT21158872	Vimukthi W.M.B.V	Borrowing Management

Group Project Test Cases

Project: Web-Based Library Management System

User Management

Member: Pathirana A.P.C.E | Reg No: IT21077524

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
UM-01	Register new user with valid details	Admin should be able to add a new user with all correct details	Admin is logged in	1. Go to User Management 2. Click Add User 3. Enter name, email, role, ID 4. Click Submit	Name: Kasun Perera Email: kasun@sliit.lk Role: Student ID: IT21001111	User created. Success message shown. User appears in list.	-	Pending
UM-02	Register user with blank name	System should not allow registration when name is empty	Admin is logged in	1. Click Add User 2. Leave name blank 3. Fill other fields 4. Click Submit	Name: (blank) Email: kasun@sliit.lk	Error: Name is required. User is not added.	-	Pending
UM-03	Register user with invalid email	System should reject an email without proper format	Admin is logged in	1. Click Add User 2. Enter email without @ 3. Click Submit	Email: kasunatsliit.lk	Error: Invalid email format. Form is not submitted.	-	Pending
UM-04	Register user with duplicate email	Two users should not have the same email address	kasun@sliit.lk already exists	1. Click Add User 2. Enter same email 3. Click Submit	Email: kasun@sliit.lk (already registered)	Error: Email already exists. User is not created.	-	Pending
UM-05	View details of an existing user	Admin should be able to view full profile of any user	Admin is logged in. At least one user exists.	1. Go to User List 2. Click on a user name	User: Kasun Perera ID: IT21001111	Profile page shows name, email, role, and borrowing history.	-	Pending
UM-06	Update user contact details	Admin should be able to edit and save changes to a user	Admin is logged in. User exists.	1. Open user record 2. Click Edit 3. Change phone number 4. Click Save	Old: 0771234567 New: 0712222333	Updated details saved. Success message shown.	-	Pending
UM-07	Delete user with no active borrowings	Admin should be able to delete a user with no borrowed books	User has no currently borrowed books	1. Select user from list 2. Click Delete 3. Confirm	User: IT21001112 Active loans: 0	User deleted. Removed from list. Confirmation shown.	-	Pending
UM-08	Try to delete user with active borrowings	System should not allow deleting a user who has books checked out	User has 1 borrowed book	1. Select user 2. Click Delete	User: IT21001113 Active loans: 1	Error: Cannot delete user with active borrowings.	-	Pending
UM-09	Search user by name	Admin should be able to find users by name in the search bar	Admin is logged in. Multiple users exist.	1. Go to User List 2. Type name in search box 3. Press Enter	Keyword: Pathirana	Matching records displayed. Unrelated users not shown.	-	Pending
UM-10	Student tries to access user management	Students should not be allowed to access the admin section	A student is logged in	1. Try to go to /admin/users directly	Logged in as: Student	Access denied. Redirected to home page or error page.	-	Pending

Book Management

Member: Samarasinghe P.D.P | Reg No: IT21054372

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
BM-01	Add new book with valid details	Librarian should be able to add a new book successfully	Librarian is logged in	1. Go to Book Management 2. Click Add Book 3. Fill all fields 4. Click Submit	Title: Intro to Algorithms Author: Cormen ISBN: 9780262033848 Copies: 3	Book added. Appears in list with availability = 3.	-	Pending
BM-02	Add book with duplicate ISBN	System should not allow two books with the same ISBN	Book with ISBN 9780262033848 already exists	1. Click Add Book 2. Enter same ISBN 3. Click Submit	ISBN: 9780262033848 (duplicate)	Error: A book with this ISBN already exists.	-	Pending
BM-03	Add book with empty title	Title is required and should not be left blank	Librarian is logged in	1. Click Add Book 2. Leave title empty 3. Fill other fields 4. Click Submit	Title: (blank) Author: Cormen	Error: Title is required. Form is not submitted.	-	Pending
BM-04	View book details	Users should be able to see full details of any book	At least one book exists	1. Go to Book List 2. Click on any book title	Book: Intro to Algorithms	Details page shows title, author, ISBN, category, and copies.	-	Pending
BM-05	Update number of copies	Librarian should be able to edit the number of copies	Librarian is logged in. Book exists.	1. Open book record 2. Click Edit 3. Change copies from 3 to 5 4. Click Save	Copies: 3 changed to 5	Book updated. Available copies shows 5.	-	Pending
BM-06	Delete a book not on loan	Librarian should be able to remove a book nobody has borrowed	Book is not currently borrowed	1. Select book 2. Click Delete 3. Confirm	ISBN: 9780262033848 Borrowed: No	Book removed. No longer visible in list.	-	Pending
BM-07	Try to delete a book on loan	System should prevent deletion if a book is borrowed	Book has an active borrowing record	1. Select a borrowed book 2. Click Delete	Borrowed by IT21001111	Error: Cannot delete a book that is currently on loan.	-	Pending
BM-08	Search book by title keyword	Users should be able to search by part of the title	Multiple books exist	1. Go to Book Search 2. Type keyword 3. Press Enter	Keyword: Algorithm	Books with Algorithm in the title are shown.	-	Pending
BM-09	Search book by author name	Search should filter results by author	Books with different authors exist	1. Go to search 2. Select Author filter 3. Type name	Author: Cormen	Only books by Cormen are shown.	-	Pending
BM-10	Add book with zero copies	System should not allow adding a book with zero copies	Librarian is logged in	1. Click Add Book 2. Set copies to 0 3. Click Submit	Copies: 0	Error: Number of copies must be at least 1.	-	Pending

Borrowing Management

Member: Vimukthi W.M.B.V | Reg No: IT21158872

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
BR-01	Borrow an available book	Student should be able to borrow a book that has copies available	Student is logged in. Book has at least 1 copy.	1. Search for book 2. Click Borrow 3. Confirm	Book: Clean Code Copies: 2 Student: IT21158872	Borrow recorded. Copies reduced by 1. Due date set automatically.	-	Pending
BR-02	Try to borrow book with no copies	System should not allow borrowing when no copies are available	Book has 0 copies available	1. Search for book 2. Click Borrow	Book: 0 copies left	Error: No copies available. Reserve option shown instead.	-	Pending
BR-03	View currently borrowed books	Student should be able to see all books they have borrowed	Student has at least one active borrowing	1. Go to My Borrowings 2. View active loans	Student: IT21158872	Table shows borrowed books with names and due dates.	-	Pending
BR-04	Extend borrowing period before due date	Student or librarian should be able to extend the due date	Book is borrowed and not yet overdue	1. Open borrow record 2. Click Extend 3. Confirm	Original due: 10 May 2026 Extension: 7 days	New due date set to 17 May 2026. Success message shown.	-	Pending
BR-05	Try to extend an overdue book	System should not allow extension if book is already overdue	Book due date has already passed	1. Open overdue borrow record 2. Click Extend	Due: 01 May 2026 Today: 05 May 2026	Error: Cannot extend overdue book. Return and pay fine first.	-	Pending
BR-06	Cancel borrow before pickup	Librarian should be able to cancel a borrow before student collects	Borrow record exists but not yet collected	1. Open borrow record 2. Click Cancel Borrow 3. Confirm	Borrow ID: BR2001 Status: Pending pickup	Record cancelled. Book availability restored.	-	Pending
BR-07	Student tries to borrow beyond limit	Student should not borrow more than the maximum allowed	Student has 3 borrowed books. Limit is 3.	1. Search for another book 2. Try to borrow it	Student: IT21158872 Current loans: 3 Limit: 3	Error: Borrow limit reached. Cannot borrow more books.	-	Pending
BR-08	Try to borrow same book twice	Student should not borrow two copies of the same book	Student already has Clean Code borrowed	1. Search for Clean Code 2. Click Borrow	Book: Clean Code already borrowed by same student	Error: You already have this book borrowed.	-	Pending
BR-09	Due date set correctly after borrowing	System should set due date based on lending policy (14 days)	Lending policy is 14 days	1. Borrow a book 2. Check due date on borrow record	Borrow date: 05 May 2026 Policy: 14 days	Due date set to 19 May 2026.	-	Pending
BR-10	Librarian views all active borrowings	Librarian should see all active borrow records across all students	Librarian is logged in. Multiple records exist.	1. Go to Borrowing Management 2. Click View All Active Loans	Multiple students with loans	Table shows student name, book, and due date for all active loans.	-	Pending

Return Management

Member: Weerasekara W.B.M.K.D | Reg No: IT21047756

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
RT-01	Return a book on time	Librarian should be able to record a return before due date	Librarian is logged in. Book borrowed and not yet overdue.	1. Go to Return Management 2. Search student or book 3. Click Return 4. Confirm	Student: IT21047756 Book: Clean Code Due: 19 May Return: 10 May	Return recorded. Availability increases by 1. No fine applied.	-	Pending
RT-02	Return an overdue book	System should detect overdue and calculate fine automatically	Book is overdue by 2 days. Fine rate Rs.10/day.	1. Go to Return Management 2. Select overdue record 3. Click Return	Due: 03 May 2026 Return: 05 May 2026 Rate: Rs.10/day	Return recorded. Fine of Rs.20 calculated and shown.	-	Pending
RT-03	View return history of a student	Librarian should see all past returns by a student	Student has at least one past return	1. Search for student 2. Open return history	Student: IT21047756	List shows returned books with return date and fine if applicable.	-	Pending
RT-04	Record return with no active borrow	System should not allow return if student has no active borrow	Student has no active borrow for this book	1. Search student and book 2. Try to record return	Book not borrowed by student	Error: No active borrow record found for this book.	-	Pending
RT-05	Update incorrect return record	Librarian should be able to fix a wrong return date	Return record with wrong date exists	1. Find return record 2. Click Edit 3. Update date 4. Click Save	Wrong: 01 May Correct: 04 May	Date updated. Fine recalculated if book was overdue.	-	Pending
RT-06	Delete incorrect return record	Librarian should be able to remove a record added by mistake	Incorrect return record exists	1. Select return record 2. Click Delete 3. Confirm	Return ID: RT1001 (mistake)	Record deleted. Book status changed back to borrowed.	-	Pending
RT-07	Book availability updated after return	After return, book should become available for others	Book had 0 copies when borrowed	1. Record return 2. Check book availability	Available before return: 0	Available copies becomes 1 after return.	-	Pending
RT-08	No fine for on-time return	System should not create a fine if returned on or before due date	Book returned on due date	1. Record return on due date 2. Check fine record	Due: 05 May 2026 Return: 05 May 2026	Fine = Rs.0. No fine record created.	-	Pending
RT-09	Confirmation shown after return	Confirmation message should appear after recording a return	Return recorded successfully	1. Librarian records return 2. Check for message on screen	Book: Clean Code Student: IT21047756	Confirmation: Book returned successfully on 05 May 2026.	-	Pending
RT-10	Return two books at once for same student	Librarian should process two returns for same student together	Student has 2 borrowed books	1. Select student 2. Select both borrow records 3. Click Return All	Book 1: Clean Code Book 2: Design Patterns Student: IT21047756	Both returns recorded. Availabilities updated. Fines calculated if overdue.	-	Pending

Reservation Management

Member: Rathnayake R.M.I | Reg No: IT21060144

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
RV-01	Reserve a book that is not available	Student should be able to reserve a book with no copies left	Student is logged in. Book has 0 copies.	1. Search for book 2. Click Reserve 3. Confirm	Book: Design Patterns Available: 0 Student: IT21060144	Reservation created. Student notified when book becomes available.	-	Pending
RV-02	Try to reserve an available book	System should not allow reservation if copies are in stock	Book has copies available	1. Search for available book 2. Click Reserve	Book: 2 copies available	Error: Book is available. Please borrow it directly.	-	Pending
RV-03	View existing reservations	Student should see a list of their current reservations	Student has at least one active reservation	1. Log in as student 2. Go to My Reservations	Student: IT21060144	All active reservations shown with book name, date, and status.	-	Pending
RV-04	Cancel an active reservation	Student should be able to cancel a reservation they no longer need	Student has an active reservation	1. Go to My Reservations 2. Select reservation 3. Click Cancel 4. Confirm	Reservation: Design Patterns	Reservation cancelled and removed from list.	-	Pending
RV-05	Modify reservation details	Student should be able to update an existing reservation	Active reservation exists	1. Open reservation 2. Click Edit 3. Update details 4. Click Save	Change priority: Normal to High	Reservation updated. Changes saved.	-	Pending
RV-06	Try to reserve the same book twice	System should not allow two reservations for the same book	Student already has a reservation for Design Patterns	1. Search for same book 2. Try to reserve again	Book: Design Patterns already reserved by same student	Error: You already have a reservation for this book.	-	Pending
RV-07	Student notified when book becomes available	Student should get a notification when reserved book is returned	Student has reservation. Another student returns the book.	1. Another student returns book 2. System checks queue 3. Notification sent	Book returned. Queue checked automatically.	Notification sent: Book is now available. Collect within 2 days.	-	Pending
RV-08	Reservation cancelled if not collected in time	If student does not collect in time, reservation should auto-cancel	Student notified but did not collect within 2 days	1. System checks expired reservations 2. Auto-cancel triggered	Notified: 01 May Deadline: 03 May Check: 04 May	Reservation auto-cancelled. Book made available for next in queue.	-	Pending
RV-09	Librarian views all reservations	Librarian should see the full reservation list across all students	Librarian is logged in. Reservations exist.	1. Go to Reservation Management 2. Click View All Reservations	Multiple students with reservations	Table shows reservations with student name, book, and date placed.	-	Pending
RV-10	Reservation queue is first come first served	Book should go to the student who reserved it first	Two students reserved same book at different times	1. Both reserve same book 2. Book is returned 3. Check who is notified first	Student A: 01 May Student B: 03 May	Student A notified first as they reserved earlier.	-	Pending

Fine Management

Member: Dolawatta T.Y.R | Reg No: IT21101274

TC ID	Test Title	Description	Preconditions	Test Steps	Test Data	Expected Output	Actual Output	Status
FM-01	Fine calculated correctly for overdue return	System should calculate fine based on number of late days	Book is overdue. Fine rate is Rs.10/day.	1. Student returns overdue book 2. System calculates fine automatically	Due: 01 May Return: 05 May Late: 4 days Rate: Rs.10	Fine = Rs.40 (4 x 10). Fine record created automatically.	-	Pending
FM-02	No fine for on-time return	Fine should be zero if returned on or before due date	Book returned on due date	1. Return book on due date 2. Check fine record	Due: 05 May Return: 05 May	Fine = Rs.0. No fine record created.	-	Pending
FM-03	View fine records for a student	Librarian should be able to check outstanding fines for any student	Student has an unpaid fine	1. Search for student 2. Go to Fine Records	Student: IT21101274	Fine records shown with book, late days, amount, and paid/unpaid status.	-	Pending
FM-04	Mark fine as paid	Librarian should be able to update a fine to paid after receiving payment	Fine exists with status Unpaid	1. Open fine record 2. Click Mark as Paid 3. Confirm	Fine: Rs.40 Status: Unpaid	Fine status changes to Paid. Date and time of payment recorded.	-	Pending
FM-05	Delete an incorrect fine record	Librarian should be able to remove a fine added by mistake	An incorrect fine record exists	1. Find wrong fine record 2. Click Delete 3. Confirm	Fine ID: FM2001 (entered by mistake)	Fine record deleted. Student balance updated.	-	Pending
FM-06	Student with unpaid fine cannot borrow	System should block borrowing if student has an outstanding fine	Student has unpaid fine of Rs.40	1. Student tries to borrow a book	Student: IT21101274 Unpaid fine: Rs.40	Error: You have a fine of Rs.40. Please pay before borrowing.	-	Pending
FM-07	Multiple fines for same student	System should handle more than one fine for the same student	Student returns 2 overdue books at same time	1. Return book 1 (3 days late) 2. Return book 2 (5 days late) 3. Check fine records	Book 1: Rs.30 Book 2: Rs.50	Two separate fine records created. Total = Rs.80.	-	Pending
FM-08	Add manual fine for damaged book	Librarian should be able to add a custom fine for book damage	Librarian is logged in. Book returned damaged.	1. Go to Fine Management 2. Click Add Fine 3. Select student and book 4. Enter amount and reason 5. Click Save	Reason: Book damaged Amount: Rs.500 Student: IT21101274	Fine of Rs.500 added with reason. Shown in student fine records.	-	Pending
FM-09	Generate fine report	Librarian should be able to view a report of all outstanding fines	Multiple unpaid fines exist	1. Go to Fine Management 2. Click View Report	Date range: May 2026	Report shows all unpaid fines with student name, amount, and total.	-	Pending
FM-10	Updated fine rate only affects new returns	If admin changes fine rate, old records should not change	Admin changes rate from Rs.10/day to Rs.15/day	1. Admin updates rate to Rs.15 2. Student returns book 2 days late	Old rate: Rs.10 New rate: Rs.15 Late: 2 days	Fine = Rs.30 using new rate. Old records are not affected.	-	Pending